

# ELI — Perception

ELI v2.4.82

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## ELI Perception — Vision, Voice, Wake Word, Tone, OS Control

eli/perception/ (~20 files). ELI's senses and hands: local vision, speech-to-text, text-to-speech, a **self-trained wake-word detector**, a **voice-profile/tone** subsystem, and OS control. All local, no APIs, no third-party accounts.

### Files

| File            | LOC   | Role                                                                                                               |
|-----------------|-------|--------------------------------------------------------------------------------------------------------------------|
| mic_resolver.py | ~580  | <b>cross-platform mic auto-resolve</b> — ranked PortAudio candidates, isolated live probes, Linux PULSE_SOURCE pin |
| audio_stt.py    | ~1.6k | STT + mic capture + ducking + adaptive pause + wake/voice capture                                                  |

| File                           | LOC   | Role                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| wakeword.py                    | ~450  | <b>self-trained, music-robust wake-word detector</b><br>(openWakeWord features + custom head)                                                                                                                                                                                                                                          |
| voice_profile.py               | ~420  | <b>prosody + labelled-emotion</b><br>(tone/question detection)                                                                                                                                                                                                                                                                         |
| tts_router.py                  | 1219  | Piper/pyttsx3/espeak router + char:/clone:/natural: voice resolution;<br>neural_fallback_state() makes a neural→Piper fallback observable                                                                                                                                                                                              |
| voice_fx.py                    | 252   | <b>voice styles</b> (calm / robotic / energetic / synthetic / refined) — base voice + ffmpeg effect chain; user preset store                                                                                                                                                                                                           |
| tts_xtts.py                    | 394   | <b>voice cloning</b> from a reference sample (Coqui XTTS-v2).<br>Zero-shot: add_clone() only registers a reference clip, conditioning happens at synthesis. Bundled in the Linux AppImage from v2.1.65; when absent the voice still registers and synthesis falls back to Piper — <b>loudly</b> , not silently, which was a live fault |
| vision.py                      | 698   | locate UI elements on screen (AT-SPI → local VL-ground → OCR)                                                                                                                                                                                                                                                                          |
| os_controller.py               | 572   |                                                                                                                                                                                                                                                                                                                                        |
| screen_locator.py              | 497   |                                                                                                                                                                                                                                                                                                                                        |
| screen_analysis.py             | ~180  | depth modes, prior-screen memory, research-context prompts                                                                                                                                                                                                                                                                             |
| ui_ground.py                   | ~320  | local GGUF precision click + computer-use agent (no HTTP)                                                                                                                                                                                                                                                                              |
| desktop_capabilities.py        | ~170  | cross-OS<br>input/screenshot/locate probe; feeds the awareness briefing and SELF_TEST                                                                                                                                                                                                                                                  |
| media_deps.py                  | ~140  | bundled-python + PATH discovery for yt-dlp/mpv/desktop CLIs                                                                                                                                                                                                                                                                            |
| gaze_engine.py                 | 358   | log housekeeping<br>file-type analysers<br>periodic screen glances (off by default)                                                                                                                                                                                                                                                    |
| log_rotation.py                | 224   |                                                                                                                                                                                                                                                                                                                                        |
| analyze_pdfs/image/mesh/csv.py | ~600  |                                                                                                                                                                                                                                                                                                                                        |
| ambient_vision.py              | 238   |                                                                                                                                                                                                                                                                                                                                        |
| local_whisper_stt.py           | 335   | streaming voice worker                                                                                                                                                                                                                                                                                                                 |
| voice_worker_streaming.py      | small |                                                                                                                                                                                                                                                                                                                                        |

## Vision (vision.py + ambient\_vision.py + analyze\_image.py)

The working local-vision stack (see memory eli-image-analysis): - VL model (default Qwen2.5-VL, configurable) loaded via Qwen25VLChatHandler; **CPU clip forced** by monkeypatching mtmd\_cpp.mtmd\_init\_from\_file(use\_gpu) — the GPU mtmd clip path **segfaults** on compute-7.5 cards. - **Hot-swap**: unload text model → load VL → infer → restore in finally, holding gguf\_inference.\_LLM\_CALL\_LOCK. - **Co-resident fast path**: a small model (Moondream2 Q4) loaded first, with the text model's ctx capped (vision\_coresident\_text\_ctx) so both fit 8GB; prefer\_fast=True uses it without a swap. - Images downsampled to vision\_max\_image\_px (1280) to avoid context overflow; repeat\_penalty/top\_k/top\_p set to kill a repetition loop. - ambient\_vision.py: optional periodic screen glances (OFF by default). A guarded daemon re-reads the toggle/interval each cycle and **skips a glance whenever the shared LLM lock is busy** (so it never steals the model mid-reply). Stores a short description as memory for rolling awareness.

## Screen locate & click (screen\_locator.py + ui\_ground.py)

Three-strategy chain for “find/click X on screen” — all local:

1. **AT-SPI** (Linux) — real widget tree: role, bounds, invoke action. Best for labelled buttons and accessible apps.
2. **Local VL-ground** (optional) — ui\_ground\_backend=auto|local\_gguf runs a local GGUF (Phi-Ground / OS-Atlas / UGround naming in models/) or the primary VL with a JSON box prompt. For icon-only or non-text UI.
3. **OCR** — Tesseract word boxes + fuzzy match; click centre coordinate.

**Computer-use agent** (computer\_use\_backend=auto|local\_gguf) — optional UI-TARS-style GGUF in models/ for multi-step screenshot→action loops. Off by default.

Configure on GUI **Screen tab** or settings.json: ui\_ground\_backend, computer\_use\_backend, screen\_analysis\_depth (quick|standard|deep).

## Deep screen analysis (screen\_analysis.py + SCREEN\_READ\_ANALYZE)

Voice examples that route to a real screenshot + VL + OCR (+ text-model fusion):

- “What’s on my screen?” — standard depth
- “Analyze my screen in depth / exactly what’s on my screen” — deep audit
- “How does this relate to my research?” — pulls local memory + KG context
- “Do you remember seeing this in previous sessions?” — prior screen\_awareness glances

No network. Missing OS tools (mpv, tesseract, ydotool) are offered via grounded\_remediation **after** a failed attempt — never as a pre-install gate.

## Cross-OS desktop input (desktop\_capabilities.py + platform\_compat.py)

| OS             | Mouse/keyboard                                   | Screenshot       |
|----------------|--------------------------------------------------|------------------|
| Linux X11      | xdotool → PyAutoGUI                              | scrot            |
| Linux Wayland  | ydotool+ydotool → xdotool (XWayland) → PyAutoGUI | grim+slurp       |
| Windows        | PyAutoGUI                                        | Pillow ImageGrab |
| macOS          | PyAutoGUI (+ Accessibility permission)           | screencapture    |
| Android/Termux | intents only                                     | N/A              |

See docs/CROSS\_PLATFORM.md for the full matrix and install commands per OS.

## Speech-to-text (audio\_stt.py, local\_whisper\_stt.py, mic\_resolver.py)

**Microphone selection (mic\_resolver.py, v2.3.44).** On every OS, ELI builds a ranked list of PortAudio inputs — USB and headset/Bluetooth (AirPods, Turtle Beach Chat, Trust USB, ...) before built-in, webcam, and virtual loopback routes — and probes each in an isolated subprocess at the device's native sample rate until one delivers live audio. Linux additionally tries PipeWire/Pulse sources with a process-local PULSE\_SOURCE pin after direct hardware. Honours ELI\_MIC\_DEVICE\_INDEX (explicit override) and ELI\_MIC\_AUTORESOLVE=0 (OS default only). Diagnostic: `python -m eli.tools.mic_diag`.

A large, feature-dense STT module: - Mic capture + Whisper transcription (local\_whisper\_stt). - **Output ducking** — lowers system sink volume while listening (`_eli_duck_output/_eli_restore_output` via `wpctl`) so ELI doesn't hear itself. - **Echo/self-hearing suppression** (`_eli_echo_like_assistant_output, _eli_media_probably_audible`). - Transcript cleanup: `_cleanup, _collapse_repeated_phrase, _eli_fast_command_alias` (maps spoken phrases to commands), `_is_safe_direct/_allow_direct_chat` (gates which utterances bypass to chat). - ALSA stderr suppression to keep the console clean. - **Duration-adaptive end-of-phrase pause** (`_listen_adaptive_pause, 2026-06-08`) — stock `sr.listen()` reads `pause_threshold` once per phrase, so one value can't be both snappy for commands and tolerant of long dictation. A faithful copy of `sr`'s capture with a single dynamic condition: short commands finalise after `ELI_STT_SHORT_PAUSE` (0.5s) of silence; a prompt speaking past `ELI_STT_LONG_AFTER` (12s) needs `ELI_STT_LONG_PAUSE` (2s), so a mid-sentence pause no longer cuts it. Flag-gated (`ELI_STT_ADAPTIVE_PAUSE`) with fallback to stock `listen()`. - **Generic mic-capture script** — one mechanism (no second microphone) drives both wake-word enrollment and voice/emotion training: a list of {prompt, sink} steps, each captured (past the TTS-echo guards) clip routed to its sink, the next cue spoken, a done callback after the last. `begin_capture_script / begin_wake_enrollment / begin_voice_training`. - **Acoustic wake hook** — when unarmed and a wake model is trained, the captured audio is scored by `wakeword` and, if it fires, the wake word is injected so the existing `VoiceGate` arms — catching the wake word even when whisper transcribed the music. **Per-turn tone hook** — on a real command, `voice_profile.classify_tone` is run and published on a side-channel for cognition (below).

## Wake word (wakeword.py, 2026-06-08) — self-trained, robust over music

Transcription-based wake matching can't hear "computer" over loud music (whisper transcribes the music). Instead ELI **trains its own** detector, 100% locally: - **Positives**: the wake phrase synthesised by ELI's OWN **Piper TTS** across several voices and speeds. - **Augmentation (the robustness)**: each positive is mixed with noise/music at random SNRs, so the classifier learns to spot the wake word *through* a music bed. - **Features**: `openWakeWord`'s open, bundled `melspectrogram`→embedding extractor (no account, no download barrier). - **Custom head**: a small torch classifier ELI owns; `train_model()` → `models/wakeword/eli_wake_head.pt` (gitignored). `WakeDetector.score_audio/is_wake` slide a 1.5s window. Validated: clean wake = 1.00, wake+loud music (3 dB) = 1.00, hard-negative/music-only = 0.00. - **User-settable phrase** — `get/set_wake_phrases` persists any phrase ("change the wake word to athena"); it feeds both this model and the transcription matcher. - **Personalisation** — enrolled real-mic clips of the user are folded in as heavily-weighted positives. - Actions: `WAKE_SET / WAKE_TRAIN / WAKE_ENROLL`. Fully fallback-safe (`ELI_WAKE_ACOUSTIC=0`; no model → the transcription matcher).

## Voice profile + tone (voice\_profile.py, 2026-06-08) — foundation for emotion

Deliberately SEPARATE from the wake word — this is *how* the user speaks, the basis for tone/emotion and question-vs-statement: - **Prosody (real, numpy only)**: per-clip autocorrelation **F0/pitch** track, energy, voiced ratio, speaking rate, and a **terminal-pitch slope** (`analyze_prosody`). - **Question vs statement** (working): rising terminal pitch ⇒ question (`question_or_statement`). - **Labelled emotion**: `add_labelled_sample` + a nearest-centroid classifier (`train_emotion_classifier`) over neutral/happy/angry/excited/sad — robust for the small data a quick enrollment yields, upgradeable to a NN without touching callers. - **classify\_tone** returns emotion + confidence, arousal, and the question/statement cue. `build_profile` learns the user's baseline so tone is scored *relative* to how they normally sound. - **TRAIN\_VOICE** runs one unified guided session (wake reps + the same line said in each emotion); modules stay separate, the user does one flow. - **Wired into cognition**: STT publishes the

per-turn read on a fresh, timestamped side-channel (`set_last_tone`); `engine._build_enhanced_system` reads `get_last_tone` and prepends a speaker-voice cue so ELI adapts its delivery (warmer if upset, brisk if energetic). `ELI_VOICE_TONE=0` disables. Complementary to the text-based cognition/tone\_analyzer (persona preferences), not a duplicate.

## Tone / emotion adaptor (cognition/tone\_adaptor.py + cognition/emotion\_palette.py)

Decides the emotion ELI **expresses** and drives three output channels at once.

- **Two identifiers** (the “2 systems”): *acoustic* (`voice_profile.get_last_tone` — the user’s voice this turn) and *semantic* (`detect_text_emotion` — the content of what they said). `_fuse()` combines them (agreement boosts confidence; voice slightly favoured).
- **Empathetic response policy**, not mimicry: a distressed user → tender, heat → calm, playful → playful. An explicit override (`set_tone("be comedic"/"talk street")`) via `SET_TONE/CLEAR_TONE` wins outright and persists.
- **Extensible palette** (`emotion_palette.py`): 20 built-in tones (comedic, professional, deadpan, street\_smart, gremlin, tender, ...), each carrying a persona directive, Piper prosody profile, and avatar expression; users add their own in `config/voices/tones.json`.
- **Three output channels**: text (a directive folded into `_build_enhanced_system`), voice (`tts_router._piper_prosody` overlays the tone’s pace/expressiveness/pauses **and** `_apply_tone_pitch` shifts the actual audio pitch so sad sounds lower, ecstatic higher — applied on the Piper, natural-neural and clone paths), and face (`world/avatar/persona_mapper` + the animated `gui/widgets/eli_face.py` take the expression).
- **Core personality is never overwritten** — the directive explicitly shades *how* ELI delivers, not *who* he is (emergent-voice principle). `ELI_TONE_ADAPT=0` disables.

## Animated face liveness (gui/widgets/eli\_face.py + cognition/expression\_state.py)

The procedural face reacts in real time to what ELI is *doing*, via three cheap flags in `expression_state` (priority: thinking > speaking > tone): - **Lip-sync** — the mouth animates while TTS plays (`tts_router._run_tts` sets the speaking flag; amplitude-driven when available, else a natural talk oscillation). - **Thinking look** — a focused/reflective face while the model generates (the GUI sets thinking around `is_generating`). - Otherwise it shows the current expressed tone. Pure-Qt, embedded in the world panel.

## Emotional memory + proactive check-in (cognition/emotion\_timeline.py, v2.1.31)

`tone_adaptor` shades the *current* reply and then forgets: the acoustic read lives in a 12-second slot, the semantic read is recomputed from the last utterance. That makes ELI **reactive**. The emotion timeline is the durable record that makes it **proactive**.

- **Persistence** — every fused read (neutral included, so the baseline isn’t built only from bad turns) is appended to `emotion_events` in `user.sqlite3`, with the utterance that produced it, the arousal, and **the action ELI ran on the preceding turn**.
- **assess()** returns EVIDENCE, never a phrase: the sustained run and its dominant emotion, the neutral→negative transition, the ELI action at the turn the mood turned, and whether it is *unusual for this user* — measured against their own baseline, so a naturally reserved person is never read as upset.
- **Gates** (all must pass before ELI raises anything): a run of N reads rather than a single spike, per-read confidence above a floor, a credible baseline, and a cooldown so ELI cannot nag. `assess()` reports its reason either way, so a *non-check-in* is explainable too.
- **Two surfacing paths**. In-conversation, `evidence_block()` is folded into the system prompt and **the model decides whether and how to raise it** — the wording is ELI’s, never a template (see [ELI No Hard-Coded Responses]). Unprompted, the proactive daemon runs the same assessment on its tick, LLM-synthesises an opener from the evidence, and pushes an `emotion_checkin` onto

the suggestion queue (+ a pending proposal so a plain “yes” routes). With no resident model the daemon emits **nothing** rather than a canned line.

- **trend\_line()** runs every turn regardless, giving ELI continuity on how the conversation has been feeling.
- **User control** — Settings ► Cognition ► *Emotional awareness*: master toggle, exchanges before a mood counts, minimum confidence, check-in cooldown, recent-mood window, baseline history. All read at call time (no restart), all defaulting to the shipped constants. A *Tone of voice* dropdown in the same tab pins the register ELI answers in (Auto + the 20 palette tones) via `tone_adaptor.set_tone`. Kill switch: `ELI_EMOTION_MEMORY=0`.

## Text-to-speech (tts\_router.py)

Multi-backend router with graceful fallback: **XTTS-v2 natural neural** (opt-in) → **Piper** (packaged binary under `tts_piper/` or `models/tts/piper`) → **pyttsx3** → **espeak-ng / espeak**. Resolves voices from several candidate dirs so a packaged build or a source checkout both work.

*Natural neural voice* (`natural:<id>`, `tts_xtts.py`). XTTS-v2 driven by its **built-in studio speakers — no clone/reference needed** — is the human-like, expressive, punctuation-aware path. `list_natural_voices()/synthesize_natural_wav()` are cheap to probe (never load the ~1.8GB model just to enumerate) and **fall back to Piper** when the neural extra isn’t installed, so a `natural:` voice never goes silent. It is selectable at the **startup model picker / first-boot wizard** (“Voice engine: Piper vs Natural”) — the choice is persisted *before* hardware tuning so its VRAM is a deliberate up-front decision (`ELI_XTTS_DEVICE` pins the device); low-VRAM boxes keep fast Piper. Enable with `pip install -e ".[natural]"` (from the ELI project root).

*Expressive Piper prosody*. Piper at bare defaults sounds flat and clips full stops. The router now passes tuned `--sentence_silence / --length_scale / --noise_scale / --noise_w` (`_piper_prosody_args()`, env-overrideable, `ELI_TTS_PROSODY=0` to disable) so `?/!/.` land as intonation and pauses — the “Amy sounds flat” fix.

*No robotic voices*. `list_voices()` hides the espeak sys: voices and Piper low/x\_low tiers by default (they sound synthetic), surfacing them only as a genuine last resort when the box has nothing better, or when `ELI_TTS_ALLOW_ROBOTIC=1`.

**Voice library** (`runtime/voice_assets.py`) — the shipped pack is a starting point, not the limit. The module fetches the upstream Piper index (**166 voices across 45 languages, 38 of them English** — US/GB, Scottish, Northern and Southern English) and exposes it through `list_available_voices()` with per-voice presence, download size and licence, cached on disk so the picker still works offline. `download_voice()` streams from the index’s exact paths and **verifies the published md5** before the file is put in place, so a truncated or proxy-mangled download is rejected rather than surfacing later as a corrupt-model crash. The GUI path is Settings > VOICE / TTS > *Get more voices / accents...* (`gui/widgets/voice_downloader.py`); every voice dropdown repopulates on install.

*Redistribution policy*. `RESTRICTED_VOICE_NAMES` (ryan — CC-BY-NC-SA; lessac — Blizzard/Lessac, not cleared; cori — pending) are never put in a release asset. The rule is keyed on the **voice name, so every quality variant is covered**, and `scripts/asset_release_policy.py` imports it so the build and the runtime cannot drift. The user may still download them for personal use; the dialog states this rather than hiding the voice.

*Config integrity*. A `.onnx` without its `.onnx.json` cannot be loaded by Piper, so it is invisible to `list_voices()` while still occupying ~60 MB. `incomplete_voices()` finds these and `repair_voice_configs()` fetches just the missing ~5 KB config. `tests/test_voice_library.py` guards against shipping the broken state again.

**Style presets** (`voice_fx.py`, `char:calm`, `char:robotic`, ...) are a redistributable Piper base voice plus an ffmpeg effect chain (pitch / speed / filters). They are **generic sound styles**, not bundled impersonations of copyrighted characters. Ideal Piper bases for some historical presets (lessac, joe, ryan) are excluded from the public bundle over dataset licensing; each preset carries an **ordered, gender-matched fallback chain** against installed voices only. **Voice cloning** (`tts_xtts.py`, `clone:<name>`) reproduces a voice

from a short reference clip via Coqui XTTS-v2 — an opt-in extra (eli-v2.0[clone]) that falls back to a normal voice when the model isn't installed rather than going silent. Both char: and clone: resolve through synthesizer\_wav (browser voice) and \_run\_tts (host playback), and appear in the GUI voice picker.

## OS control (os\_controller.py)

**Platform-aware** (Linux/Windows/macOS) screenshot, volume, keyboard, clipboard. take\_screenshot(region) picks the right tool per platform (gnome-screenshot, PIL ImageGrab, platform CLIs); fails gracefully where a capability isn't available (e.g. area screenshots on some Windows installs). screen\_locator.py finds UI targets for click/automation.

## File analysers

analyze\_pdfs, analyze\_image, analyze\_csv — typed handlers behind the ANALYZE\_\* actions, feeding extracted content (text/OCR/structure) into the grounded pipeline.

## Honest assessment

- **Strong:** this is the layer that makes ELI genuinely multimodal and local — eyes (VL + OCR + ambient), ears (Whisper + ducking), mouth (Piper/espeak), hands (cross-platform OS control). The vision co-residence + hot-swap + CPU-clip workaround is real engineering against hard 8GB-GPU constraints.
- **Weak / watch:**
  1. **Vision latency** — the hot-swap unloads/reloads the text model per glance; even the co-resident path runs CPU clip (~3.5s). Acceptable, not fast.
  2. audio\_stt.py is a **2.5k-line** module mixing capture, ducking, echo suppression, command aliasing, and cleanup — wants splitting.
  3. **Residual “7B” comment** in ambient\_vision.py (“the 7B vision model hot-swaps...”) — cosmetic, but inconsistent with the model-agnostic line; should read “the vision model”. (Flagged for a future scrub.)
  4. gaze\_engine.py is experimental and off by default — fine, but it's carried weight.
  5. Platform tools are detected at call time; a machine missing gnome-screenshot/wpctl/piper degrades silently to fallbacks (good) but there's no single “what perception capabilities do I actually have here?” probe surfaced to the user.

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## Update — 2026-06-09 (VRAM-aware STT; drag-drop keeps the full path)

- **STT is VRAM-aware** (local\_whisper\_stt). faster-whisper defaults to GPU only when the card is big enough for whisper AND a typical main model (≥12 GB total VRAM), else CPU — so on an 8 GB card whisper no longer preloads ~2 GB and starves the main model (observed: free\_vram 4083 MB → gpu\_layers=11; after the fix free\_vram 7092 MB → gpu\_layers=99). small.en int8 on CPU is ~1-2 s for a short command. Explicit ELI\_WHISPER\_DEVICE still wins; ELI\_WHISPER\_GPU\_MIN\_MB tunes the threshold.
- **Drag-and-drop keeps the FULL path.** A file dropped into the chat input now expands to [File: <full path>] followed by the inlined content (was content + basename only), so “fix/examine/edit this file” can resolve the file on disk while “summarise this” still has the content. PDFs keep their path too; combined with the FIX\_FILE last-file memory, a bare follow-up “fix it” recovers the file named a turn earlier.